

Machine Learning Libraries Developers Must Know

Library	Language	Purpose / Use-Case	Key Features
TensorFlow	Python, C++	Deep Learning, Neural Networks	Scalable, production-ready, supports distributed training
PyTorch	Python, C++	Deep Learning, Research	Dynamic computation graphs, easy experimentation
Scikit-Learn	Python	Classical ML algorithms	Easy API, regression, classification, clustering
Keras	Python	High-level neural networks	User friendly, runs on TensorFlow backend
XGBoost	Python, R, C++	Gradient boosting, tabular data	High performance, widely used in competitions
LightGBM	Python, R	Gradient boosting by Microsoft	Faster training, handles large datasets
CatBoost	Python, R	Gradient boosting for categorical data	No need for one-hot encoding, extremely fast
Pandas	Python	Data manipulation	Dataframes, cleaning, transformations
NumPy	Python	Numerical computing	Fast arrays, foundational for ML stack
OpenCV	C++, Python	Computer Vision	Image/video processing, CV algorithms
NLTK	Python	Natural Language Processing	Tokenization, tagging, stemming, parsing
spaCy	Python	Industrial NLP	Fast, production-level NLP pipelines